

The schematic diagram illustrates the internal wiring of a USB Type-C to USB-A adapter. On the left, the USB Type-C connector (J1) is shown with pins for VBUS, CC1, CC2, D+, D-, and GND. The USB-A connector (U1) is on the right, with pins for VBUS, D+, D-, and GND. A USB Type-B connector (U2) is at the bottom, connected to the USB-A connector via a cable. The schematic includes components like resistors R1 and R2 (5.1k) connected to the CC pins, a diode D1 (SMF5.0A) connected to the D+ pin, and a USB Type-B connector (U2) connected to the USB-A connector via a cable. The USB Type-C connector is connected to the USB-A connector via a cable.

The circuit diagram illustrates a USB-C to USB-A adapter. It features a TP5563Z01DODCR DC-DC converter (U2) which steps down the +3V3 input from the USB-C port to the +5V output required by the USB-A port. The input side includes a 2.2μH inductor (L1), a 100nF capacitor (C3), and a switch (SW). The output side has a 10μF/25V capacitor (C1), a 100nF capacitor (C2), and two 22μF capacitors (C4 and C5). Resistor R3 (33.2k) connects the +3V3 line to ground, while R4 (10k) connects the +5V line to ground. The schematic also shows various pins like VIN, EN, SWB, VBST, FB, and VBUS, along with their respective connections to ground or other components.

$V_{out} = 0.768 \text{ V} \times (1 + R3/R4) = 0.768 \times (1 + 33.2\text{k}/10\text{k}) = 3.32 \text{ V}$. EN tied to VIN; on with VBUS.
Load: GWS1670 at 28.5 dBm, about 3–4 W in Tx bursts = -1.2 A @ 3.3 V = -0.9 A from VBUS.
That is more than a 500 mA USB 2.0 port guarantees; use a USB 3 / USB-C host port or a powered hub.
Only C1 + C2 (10 μF) sit on VBUS; the USB inrush limit; the bulk is after the soft-started buck.
L1 2.2 μH (T_p typical for 3.3 V); with the socket bulk cap this rail has ~47 f effective,
and 2.2 μF keeps the LC pole inside TI Table 2 [20–68 nF up to 4.7 μH].
Layout: C1/C2 tight to VIN-GND. SW copper short. L1 and C4/C5 close. FB trace away from SW/L1.

The diagram illustrates the pin configuration and power supply for the GW16170 module. The module is shown as a yellow rectangle with pins numbered 1 to 44. The connections are as follows:

- Pin 1:** GND
- Pin 2:** +3V3
- Pin 3:** USB_D+
- Pin 4:** USB_D-
- Pin 5:** RESERVED
- Pin 6:** RESERVED
- Pin 7:** RESERVED
- Pin 8:** RESERVED
- Pin 9:** RESERVED
- Pin 10:** RESERVED
- Pin 11:** RESERVED
- Pin 12:** RESERVED
- Pin 13:** RESERVED
- Pin 14:** RESERVED
- Pin 15:** RESERVED
- Pin 16:** RESERVED
- Pin 17:** RESERVED
- Pin 18:** RESERVED
- Pin 19:** RESERVED
- Pin 20:** RESERVED
- Pin 21:** RESERVED
- Pin 22:** RESERVED
- Pin 23:** RESERVED
- Pin 24:** RESERVED
- Pin 25:** RESERVED
- Pin 26:** RESERVED
- Pin 27:** RESERVED
- Pin 28:** RESERVED
- Pin 29:** RESERVED
- Pin 30:** RESERVED
- Pin 31:** RESERVED
- Pin 32:** RESERVED
- Pin 33:** RESERVED
- Pin 34:** RESERVED
- Pin 35:** RESERVED
- Pin 36:** RESERVED
- Pin 37:** RESERVED
- Pin 38:** RESERVED
- Pin 39:** RESERVED
- Pin 40:** RESERVED
- Pin 41:** RESERVED
- Pin 42:** RESERVED
- Pin 43:** RESERVED
- Pin 44:** RESERVED

The power supply circuitry includes a +3V3 input, a 100uF capacitor (C6), and a 10uF capacitor (C7). The output is connected to the module's +3V3 pin. The module's GND pin is connected to the GND of the power supply circuit.

Pin 56 W_DISABLE1# = card RESET_N (200k pull-up on the card), pin 54 W_DISABLE2# = WAKE (10k pull-up on the card) both left open, so the card runs whenever +3V3 is up.
C6 covers the Tx burst edge; C8/C9 go one at pins 2/4 and one at pins 72/74.
Card is 22 x 30 x 3.5 mm with an MMCX antenna jack: no RF on this board.
HS: 2.5 mm SMT nut (LCSC C5301773), 29.75 mm from the socket centre on the peg side (card stop is 1.75 mm behind the pegs).

H1 M3 H2 M3 H3 M3 H4 M3 M2 standoff H2.5

GND

H1—H4: M3 board corners. H5: M2 standoff under the card's mounting notch. 30 mm card length (2230).

FID1
Fiducial

FID2
Fiducial

FID3
Fiducial

TP1
+3V3

TP2
VBUS

TP3
GND

FID1-3: 1 mm 16Fiducials for the
0.5 mm pitch socket; TP1-3: 1.5 mm
probe pads, outside the card outline.